

Homework 3 – Per-product Forecast with GK Clustering and Local PCA-LS Models

Subject: Intelligent Methods of Biomedical Data Research

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Homework Assignment

Task – Per-product forecast with GK clustering and local PCA-LS models

Goal. The deliverable of this homework is a **forecast** $\hat{y}$ of the output quality y for new (test) samples of the production line. Clustering and PCA are only *means* to that end:

- the line produces several different products, so a *single* global model cannot fit all of them well – we therefore split the data per product and build one local model per product;
- the products are not labelled, so we must **discover** them by *clustering* the inputs;
- within each product the inputs are maybe collinear, so each local model uses *PCA* preprocessing before linear LS.

Datasets. The homework must be carried out on *both* of the following `.mat` files:

- `studentData.mat` – variables `input` (687×7) and `output` (687×1): the smaller dataset, useful for fast iteration and debugging.
- `studentDataLarge.mat` – variables `inputLarge` (2060×7) and `outputLarge` (2060×1): the bit larger dataset.

In both files the rows of the input matrix are samples stored in chronological order, the 7 columns are the process inputs ($p = 7$), and the corresponding entry of the output vector is the measured output quality y . No product labels are provided – the products must be discovered by the student. The full pipeline (the three models and both evaluation splits described below) must be run *independently* on each dataset, and the results reported side-by-side so that the effect of the dataset size can be discussed.

PCA – choosing the number of components

PCA (Sec. 6.1 and 7.2 in [1]) is run on the standardised inputs and yields eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$ of the sample covariance matrix. Each λ_j equals the variance captured

by the j -th principal component, and $\sum_j \lambda_j$ is the total variance of the standardised inputs. The fraction of variance explained by the first k components is therefore

$$\text{EVR}(k) = \frac{\sum_{j=1}^k \lambda_j}{\sum_{j=1}^p \lambda_j}. \quad (1)$$

The number of retained components is the smallest k for which $\text{EVR}(k) \geq \tau$, with the threshold τ commonly chosen in $[0.90, 0.99]$. In this homework use $\tau = 0.95$, i.e. *keep enough components to explain at least 95% of the variance*. PCA may be applied either on the whole training set (one shared projection) or separately inside each cluster (one local projection per product) – this choice is left to the student, but it must be stated clearly in the report and used consistently for all clustered models.

Used algorithms. You should use GK clustering (Sec. 6.2.3 and 7.4.2 in [1]), PCA preprocessing (Sec. 6.1 and 7.2 in [1]) and linear LS (Sec. 3.1 in [1]) or better algorithms. Separate products in 3 different ways:

1. **Model A – GK on dimensions $\{2, 6\}$.** Standardise the training set and run GK with c clusters using *only* input dimensions x_2 and x_6 , then build one local PCA + LS model per cluster on the *full* input vector $\mathbf{x}$, train one local PCA + LS model per cluster (hard assignment $\arg \max_i U_{ik}$), and at test time assign each test sample to the nearest cluster using the GK Mahalanobis distance in the same two dimensions $\{2, 6\}$.
2. **Model B – no clustering (single global model).** Fit a single PCA + LS model on the entire training set and use it to forecast every test sample. This is the natural baseline for Model A.
3. **Model C – GK on all p dimensions.** Repeat the workflow of Model A but cluster on the *full* standardised input vector $\mathbf{x} \in \mathbb{R}^p$ instead of the two selected dimensions. The local PCA + LS models are built and applied exactly as in Model A.

Everytime the model needs to be trained, learn it by 2 ways - using least squares on all the training data, and using PCA to reduce the dimensionality of the input space before applying least squares.

Evaluation – two split strategies. Evaluate *all three* models with the *same two* train/test splits, with 20 % of the samples held out as the test set in both cases:

- **Chronological split:** the *last* 20 % of the samples (in the order in which they appear in dataset) are the test set, the preceding 80 % are the training set. This mimics the realistic situation where the model is trained on past production and used to forecast future samples.
- **Random split:** the test indices are drawn uniformly at random (use a fixed seed so the split is reproducible). This is the standard cross-validation style split (Sec. 8.3.2 in [1]).

All in all, you will create 24 models - each combination of clustering (A,B,C), split strategy (chronological, random), and PCA usage (with, without), and dataset (`studentData.mat`)

and `studentDataLarge.mat`). For each combination report the overall test RMSE, and analyse the results as described below.

If you will iterate on the model based on the results obtained on the data, make sure that you run your algorithms on test set only ones - otherwise the test set will become a de facto training set and the results will be optimistically biased. If you need to iterate, do it on training set using crossvalidation, or create a separate validation set from the training set and use it for iteration, keeping the test set strictly for the final evaluation.

Discussion.

- Compare the three models on each split: does clustering help, and does clustering on $\{2, 6\}$ behave differently from clustering on all p dimensions?
- Compare each model's chronological RMSE with its random RMSE: *what do you see?* If they differ, explain what this tells you about the data and what it implies for how the forecast should be validated in practice.
- Comment briefly on the role of the per-cluster PCA in the forecast quality and on your choice of where to apply PCA (whole training set vs. inside each cluster).
- Compare the results obtained on `studentData.mat` with those obtained on `studentDataLarge.mat`: do the conclusions above (effect of clustering, effect of the split strategy) change with the dataset size, and if so, how?

Submission

- Submit a zip file with the report and code according to the reporting guidelines.
- Deadline: see the online classroom.

References

- [1] O. Nelles, *Nonlinear System Identification: From Classical Approaches to Neural Networks, Fuzzy Models, and Gaussian Processes*, 2nd ed. Cham, Switzerland: Springer, 2020.